

王松宸 2024201594

1. 令 $n = 2^m$, 则 $T(n) = 2T(2^{m-1}) + 2^m$

$$\text{而 } T(2^{m-1}) = 2T(2^{m-2}) + 2^{m-1}$$

$$\therefore T(n) = 2(2T(2^{m-2}) + 2^{m-1}) + 2^m$$

$$= 2^2 T(2^{m-2}) + 2 \cdot 2^m$$

$$= 2^m T(1) + m \cdot 2^m$$

$$= n + n \lg n \geq n \lg n \quad (c=1)$$

$$\therefore T(n) = \Omega(n \lg n)$$

$$\text{又 } \because T(n) = O(n \lg n)$$

$$\therefore T(n) = \Theta(n \lg n)$$

2. $68^a = 132464 \Rightarrow a \approx 2.79513$

$$70^b = 143640 \Rightarrow b \approx 2.79512$$

$$72^c = 155424 \Rightarrow c \approx 2.79515$$

$\therefore$ 用 143640 次乘法完成
70x70 的矩阵相乘会
得到最佳的渐近运行时间

$\therefore$ Strassen 算法时间复杂度约为 $O(n^{2.81})$

$\therefore$ 该算法性能优于 Strassen 算法

3. 模仿 Strassen 算法的思路, 用更多的加法运算代替乘法运算

def fast_multiply(a, b, c, d):

$$x_1 = a(c+d)$$

$$x_2 = c(a-b)$$

$$x_3 = b(c-d)$$

$$\text{return } x_2 + x_3, x_1 - x_2$$

$$a(c+d) - c(a-b) = ad + bc$$

$$c(a-b) + b(c-d) = ac - bd$$



4. ① 根据主定理, 若 $f(n) = O(n^{\log_b a - \epsilon})$ 则 $T(n) = O(n^{\log_b a})$

此题中 $a=4$ $b=3$ $f(n) = n \lg n$

只需证 $f(n) = O(n^{\log_3 4 - \epsilon})$ 则可得 $T(n) = O(n^{\log_3 4})$

$$\text{取 } \epsilon = \frac{\log_3 4 - 1}{2} > 0$$

$$\text{此时 } \log_3 4 - \epsilon - 1 = \frac{\log_3 4 - 1}{2} > 0$$

$$\therefore \lim_{n \rightarrow \infty} \frac{\lg n}{n^{\log_3 4 - \epsilon - 1}} = 0 \Rightarrow n \lg n = O(n^{\log_3 4 - \epsilon})$$

$\therefore T(n) = O(n^{\log_3 4})$ 验证完毕

$$\textcircled{2} \text{ 令 } n = 3^m \text{ 则 } T(n) = 3T(3^{m-1}) + \frac{3^m}{m \lg 3}$$

$$\therefore \frac{T(3^m)}{3^m} = \frac{T(3^{m-1})}{3^{m-1}} + \frac{1}{m \lg 3}$$

$$\Rightarrow \frac{T(3^m)}{3^m} = \frac{1}{\lg 3} \left(1 + \frac{1}{2} + \dots + \frac{1}{m}\right) = O(\lg m)$$

$$\therefore T(n) = \frac{n}{\lg 3} O(\lg m) = O(n \lg \lg n)$$

$$\textcircled{3} f(n) = n^{\frac{5}{2}} = \Omega(n^{2+\epsilon})$$

$$\text{而 } 4 \cdot \left(\frac{n}{2}\right)^{\frac{5}{2}} \leq c \cdot n^{\frac{5}{2}} \text{ (当 } c = 4 \cdot \left(\frac{1}{2}\right)^{\frac{5}{2}} \text{ 时)}$$

$$\therefore \text{由主定理可得 } T(n) = O(f(n)) = O(n^2 \sqrt{n})$$

$$\textcircled{4} \text{ 令 } T(k) \leq ck \lg k + ck \text{ 对 } \forall k = 1, 2, \dots, n-1 \text{ 成立}$$

$$\text{则 } T(n) = 3T\left(\frac{n}{3} - 2\right) + \frac{n}{2}$$

$$\leq 3 \left[c \left(\frac{n}{3} - 2\right) \lg \left(\frac{n}{3} - 2\right) + c \left(\frac{n}{3} - 2\right) \right] + \frac{n}{2}$$

$$\leq 3 \left(c \frac{n}{3} \lg \frac{n}{3} + c \frac{n}{3} \right) + \frac{n}{2}$$

$$= cn \lg n + cn + \frac{n}{2} - c \lg 3 \cdot n$$

$$\leq cn \lg n + cn \text{ (c 足够大时成立)} \Rightarrow T(n) = O(n \lg n)$$



令 $T(k) \geq ck \lg k$ 对 $\forall k=1, 2, \dots, n-1$ 成立

$$\text{则 } T(n) = 3T(\frac{n}{3}-2) + \frac{n}{2}$$

$$\geq 3c(\frac{n}{3}-2)\lg(\frac{n}{3}-2) + \frac{n}{2}$$

$$\geq cn \lg n + \frac{n}{2} - 6c \lg(\frac{n}{3}-2) - c \cdot 1$$

$$\geq cn \lg n \quad (c \text{ 足够小时成立}) \Rightarrow T(n) = \Omega(n \lg n)$$

综上, $T(n) = \Theta(n \lg n)$

$$\textcircled{5} \text{ 令 } n = 2^m \text{ 则 } T(n) = 2T(2^{m-1}) + \frac{2^m}{m}$$

$$= 2(2T(2^{m-2}) + \frac{2^{m-1}}{m-1}) + \frac{2^m}{m}$$

$$= 2^m T(1) + 2^m (1 + \frac{1}{2} + \dots + \frac{1}{m})$$

$$= n + \Theta(n \lg \lg n)$$

$$= \Theta(n \lg \lg n)$$

$$\textcircled{6} T(n) = T(\frac{n}{2}) + T(\frac{n}{4}) + T(\frac{n}{8}) + n$$

$$= T(\frac{n}{4}) + T(\frac{n}{8}) + T(\frac{n}{16}) + \frac{n}{2} + T(\frac{n}{8}) + T(\frac{n}{16}) + T(\frac{n}{32}) + \frac{n}{4} + T(\frac{n}{16}) + T(\frac{n}{32})$$

$$+ T(\frac{n}{64}) + \frac{n}{8} + n$$

$$= \cancel{n(1 + \frac{1}{2} + \frac{1}{4} + \dots)}$$

设 $T(k) \leq ck$ 对 $\forall k=1, 2, \dots, n-1$ 成立

$$\therefore T(n) \leq c \cdot \frac{n}{2} + c \cdot \frac{n}{4} + c \cdot \frac{n}{8} + n$$

$$= c \cdot \frac{7}{8}n + n$$

$$\leq cn \quad (c > 8 \text{ 时成立}) \Rightarrow T(n) = O(n)$$

设 $T(k) \geq ck$ 对 $\forall k=1, 2, \dots, n-1$ 成立

$$\therefore T(n) \geq c \cdot \frac{7}{8}n + n$$

$$\geq cn \quad (0 < c \leq 8 \text{ 时成立}) \Rightarrow T(n) = \Omega(n)$$

$$T(n) = \Theta(n)$$





$$\begin{aligned}
 \textcircled{7} \quad T(n) &= T(n-1) + \frac{1}{n} \\
 &= T(n-2) + \frac{1}{n-1} + \frac{1}{n} \\
 &= 1 + \frac{1}{2} + \dots + \frac{1}{n} \\
 &= \textcircled{H}(\lg n)
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{8} \quad T(n) &= T(n-1) + \lg n \\
 &= \lg 1 + \lg 2 + \dots + \lg n \\
 &= \lg(n!) \\
 &= \textcircled{H}(n \lg n)
 \end{aligned}$$

用 $\frac{n}{2}$ 来拟合

$$\textcircled{9} \quad \text{分奇偶讨论} \quad T(n) = \begin{cases} \sum_{k=1}^{\frac{n}{2}} \frac{1}{\lg(2k)} & \text{偶} \\ \sum_{k=1}^{\frac{n-1}{2}} \frac{1}{\lg(2k+1)} & \text{奇} \end{cases}$$

n 为偶数时

$$\begin{aligned}
 \int_4^n \frac{dt}{\lg t} &\leq \int_1^{\frac{n}{2}} \frac{dk}{\lg(2k)} \leq \sum_{k=1}^{\frac{n}{2}} \frac{1}{\lg(2k)} \leq \frac{1}{\lg 2} + \int_1^{\frac{n}{2}} \frac{dk}{\lg(2k)} \leq \frac{1}{\lg 2} + \frac{1}{2} \int_2^{\frac{n}{2}} \frac{dt}{\lg t} \\
 &\leq \frac{1}{\lg 2} + \frac{1}{2} \int_2^8 \frac{dt}{\lg t} \\
 \int_4^n \frac{(\lg t - 1) dt}{\lg^2 t} &\leq \Rightarrow T(n) = \textcircled{H}\left(\frac{n}{\lg n}\right) \Leftarrow \frac{1}{2} \int_8^n \frac{dt}{\lg t} \\
 &\leq \frac{1}{\lg 2} + \frac{1}{2} \int_2^8 \frac{dt}{\lg t} + \int_8^n \frac{(\lg t - 1) dt}{\lg^2 t} \\
 &\leq c \frac{n}{\lg n}
 \end{aligned}$$

$$\begin{aligned}
 n \text{ 为奇数时 同理有 } \frac{n}{\lg n} &\leq \sum_{k=1}^{\frac{n-1}{2}} \frac{1}{\lg(2k+1)} \leq \frac{1}{\lg 3} + \frac{1}{2} \int_3^8 \frac{dt}{\lg t} + \frac{1}{2} \int_8^n \frac{dt}{\lg t} \leq c \frac{n}{\lg n} \\
 &\Rightarrow T(n) = \textcircled{H}\left(\frac{n}{\lg n}\right)
 \end{aligned}$$

综上, $T(n) = \textcircled{H}\left(\frac{n}{\lg n}\right)$



$$\textcircled{10} \quad T(n) = \sqrt{n} T(\sqrt{n}) + n$$

$$\Rightarrow \frac{T(n)}{n} = \frac{T(\sqrt{n})}{\sqrt{n}} + 1$$

$$\therefore \text{令 } m = \lg n \text{ 有 } \frac{T(2^m)}{2^m} = \frac{T(2^{\frac{m}{2}})}{2^{\frac{m}{2}}} + 1$$

$$\Rightarrow S(m) = S(\frac{m}{2}) + 1$$

$$\Rightarrow S(m) = \lg m = \textcircled{H}(\lg m)$$

$$\therefore \frac{T(n)}{n} = \textcircled{H}(\lg \lg n)$$

$$\Rightarrow T(n) = \textcircled{H}(n \lg \lg n)$$

5. (1) 设 $n = 2k$, 而 $k+1$ 块芯片是坏的, $k-1$ 块芯片是好的
此时两两配对则至少有一对是 1 好 1 坏的, 无法辨别

(2) 对检测设置“丢弃”策略, “都好”则丢弃一块, 其他情况则丢弃全部, 另外总数为单数则保留一块不配对。这种策略保证了 好 $\geq$ 坏 恒成立, 因此一轮检测后复杂度减半

$$(3) \text{ 由 (2) 有 } T(n) = T(\frac{n}{2}) + \frac{n}{2} \quad n^{\lg 2} \geq n^\epsilon \quad (\epsilon < 1 \text{ 时})$$

$$\text{由主定理 } T(n) = \textcircled{H}(n)$$

